



services, providing advanced insights and analytical tools.

After reviewing the soil analysis report on their phone, farmers can easily determine which nutrients are needed for each soil type, identify areas with low moisture, and pinpoint locations where water is not required.

Although the startup has filed multiple patents, it is awaiting official confirmation from the patent office. Rajul shares that patents for NutriSens have been filed in India, the USA, China, Australia, and Indonesia and are pending approval.

The startup is headquartered in Pune, with in-house manufacturing focused on sensor modifications and intellectual property-related processes. However, specific components, such as PCB fabrication, are outsourced.

The company's target audience includes farmer producer organisations, agricultural enterprises, and



Rajul Patkar, Founder and CEO, Proximal Soilsens Technologies Pvt Ltd

bio-fertiliser companies. Revenue is generated through the sale of products to these entities. While the company sells its devices, it also ensures 24/7 availability of after-sales services.

The startup faces multiple challenges, with awareness creation being

a key hurdle, as shifting farmers' mindsets can be difficult. "To address this, the team collaborates with partners already working with farmers, conducting both online and offline awareness camps. Farm visits, training sessions, and discussions on the importance of soil testing are integral to our approach," shares Rajul.

Rajul claims that their devices are now exported to Indonesia, Uzbekistan, Africa, and Thailand, contributing to the global adoption of precision farming. The team comprises seven full-time and three part-time employees, all dedicated to transforming soil testing and agricultural technology.

While discussing future plans, Rajul shares that the company has multiple products in development, including petiole analysis, water quality monitoring, and additional sensors.

—Nitisha Dubey

2 "OUR GOAL IS TO MAKE NEUROSCIENCE ACCESSIBLE THROUGH FULLY OPEN SOURCE TECHNOLOGY"

From failure to global impact—this startup is making neuroscience accessible with open source, low-cost bioamps. How did they get there?

In 2016, Deepak Khatri, inspired by a tech talk by Greg Gage, set out to build his own biopotential amplifier. He gathered components worth ₹1000, but the first attempt failed. Determined to learn, he broke down the problem, studied each element, and eventually built a working circuit for under ₹100.

By 2018, with guidance from his professor, he learned to design and manufacture PCBs. That year, he developed his first proper PCB for a biopotential amplifier (BioAmp) and launched a store on Tindie

called Upside Down Labs. The name reflected his perpetually messy workspace and a nod to Stranger Things. By December 2020, he was designing one PCB daily, refining the BioAmp EXG Pill—a compact, open source biosignal sensor. Initially developed for EMG, he soon realised it could also capture ECG and EEG, outperforming existing solutions.

Deepak's college friends, Dikshant Dahiya and Bhawna Sehgal, joined him as his interest grew. They formalised Upside Down Labs as an LLP and launched a crowdfunding



Deepak Khatri, Founder, Upside Down Labs with product

campaign, raising over \$13,000—eventually surpassing \$30,000. To date, they have sold BioAmp EXG Pills worth over \$100,000.

Today, the startup provides neuroscience kits for students and researchers and collaborates with universities and industries. It currently offers over seven different